深度学习

DEEP LEARNING-XIAOLIN HU

Part.09

# 深度学习在自然语言中的应用

# 大纲

- 典型任务
- 词表示
- 神经网络进行文本分类
- 神经网络进行机器翻译

# 自然语言处理

- **Natural language processing** (**NLP**) is a field of computer science, artificial intelligence and computational linguistics concerned with the interactions between computers and human (natural) languages, and, in particular, concerned with programming computers to fruitfully process large natural language corpora.

- Involve natural language understanding, natural language generation (frequently from formal, machine-readable logical forms), connecting language and machine perception, managing human-computer dialog systems, or some combination thereof.

From https://en.wikipedia.org/wiki/Natural_language_processing

# 典型任务

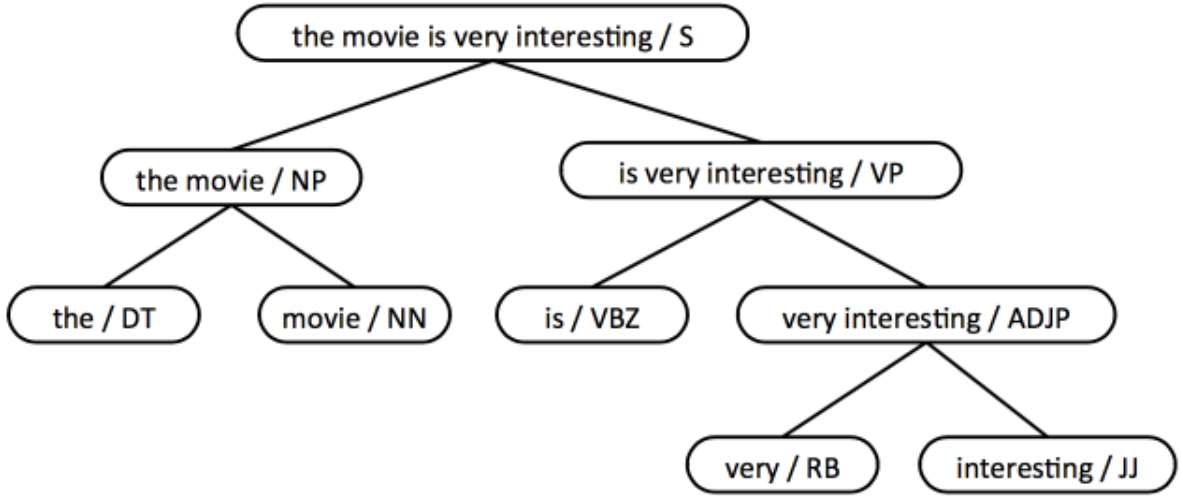
- 词性标注和句法分析
- 问答和对话系统
- 文本/文档分类
- 情感分析和观点挖掘
- 机器翻译
- 文本生成
- ……

# 词性标注和句法分析

**词性(POS)标注**

| The | movie | is | very | interesting |
|---|---|---|---|---|
| DT | NN | VBZ | RB | JJ |

**句法分析**




Slide from Prof. Minlie Huang

# 问答和对话系统

Q: How many people live in Beijing?

A: *About* ***22 million****.*

Q: What creates a cloud

A: *in meteorology , a cloud is* ***a visible mass of liquid droplets*** *or frozen crystals made of water or various chemicals suspended in the atmosphere above the surface of a planetary body.*

| | |
|---|---|
| $x_0$ | *good morning* |
| $y_0$ | *hello what can i help you with today* |
| $x_1$ | *can you book a table in a cheap price range in london* |
| $y_1$ | *i'm on it* |
| $x_2$ | *<SILENCE>* |
| $y_2$ | *any preference on a type of cuisine* |
| $x_3$ | *with french food* |
| $y_3$ | *how many people would be in your party* |
| $x_4$ | *for four please* |
| $y_4$ | *ok let me look into some options for you* |
| $x_5$ | *<SILENCE>* |
| $y_5$ | *api_call french london four cheap* |

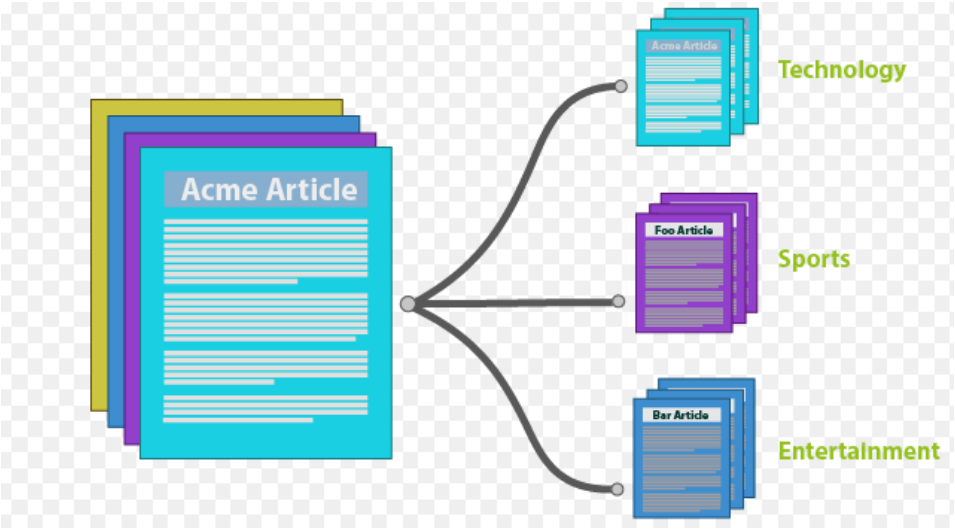
Slide from Prof. Minlie Huang

# 文本/文档分类

- 情感分类

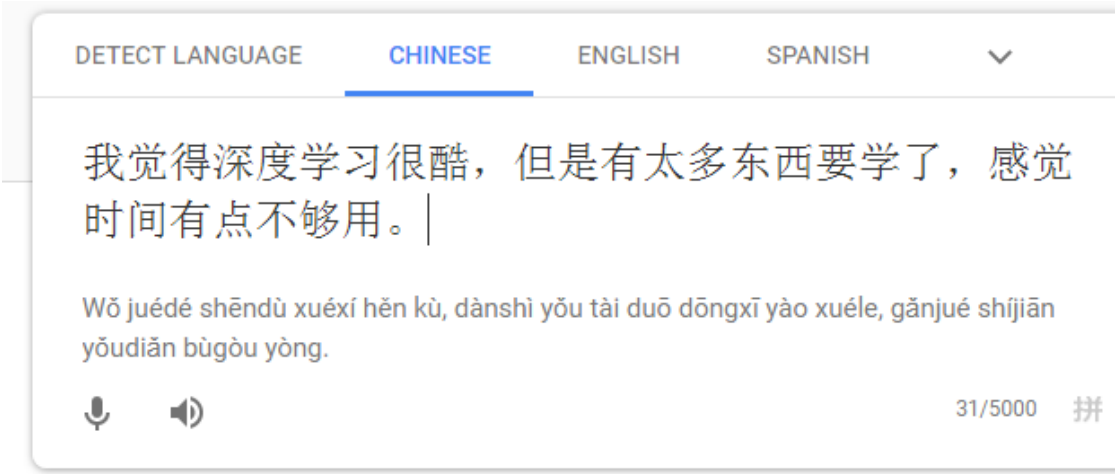
| | |
|---|---|
| *Deep learning is so cool.* | Positive |
| *This skirt looks ugly!* | Negative |
| *I attended the lecture today.* | Neutral |

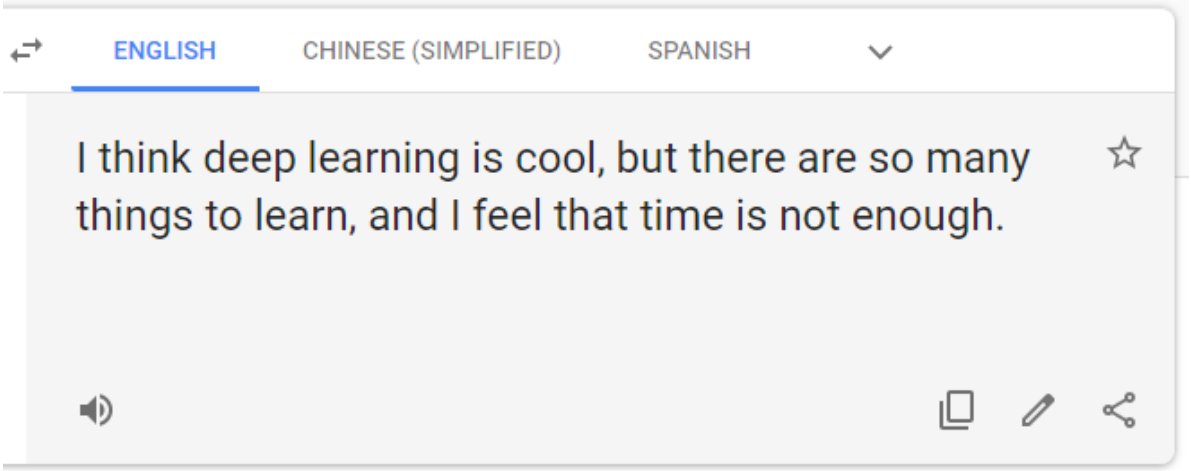
- 文档分类



# 机器翻译

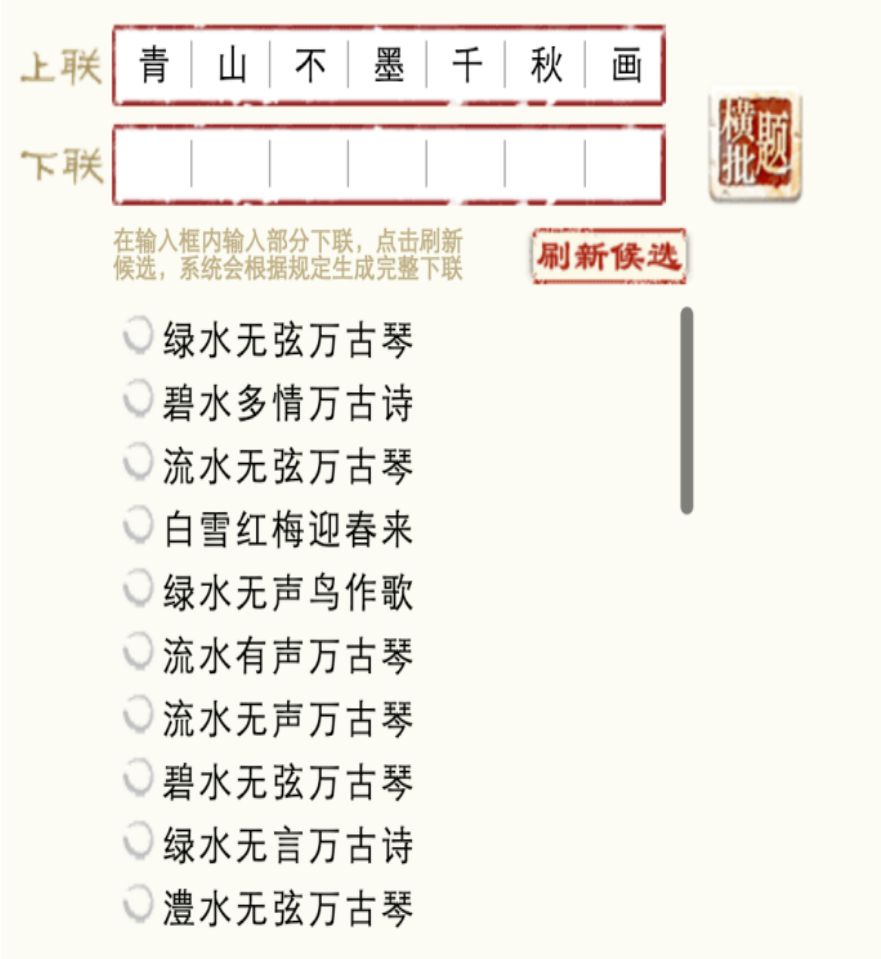
https://translate.google.com

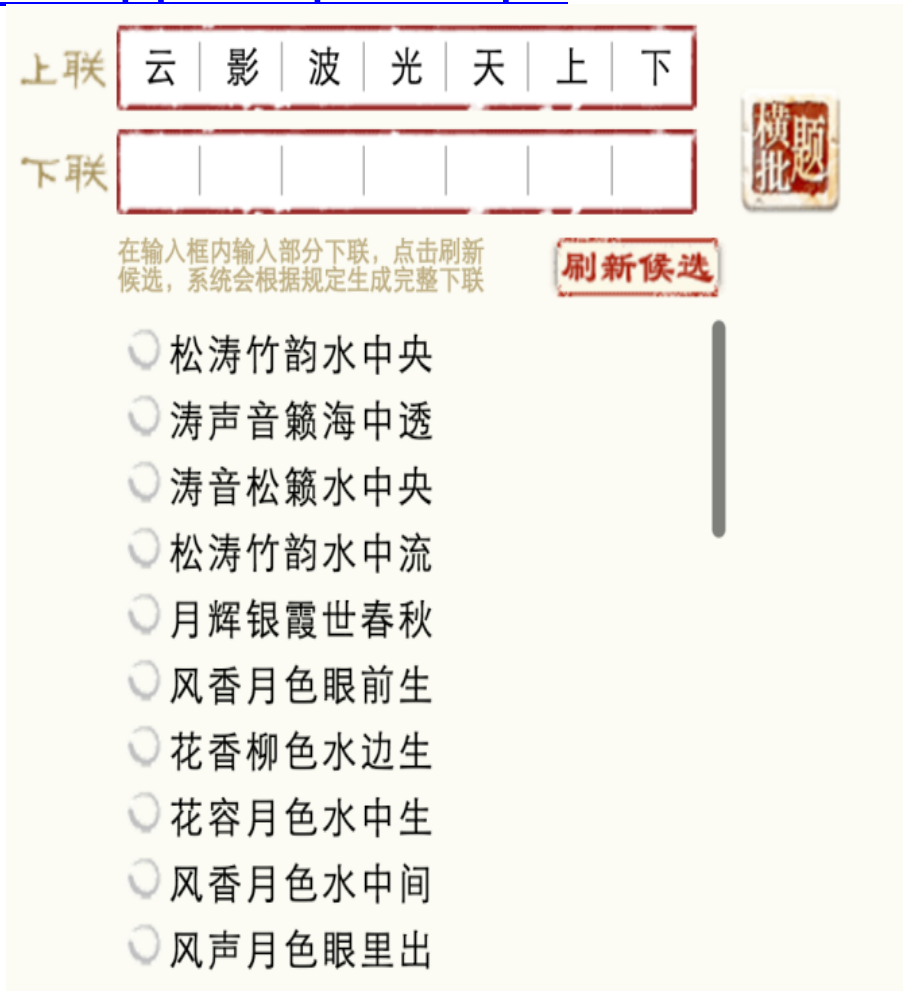


# 中国对联生成

http://couplet.msra.cn/app/couplet.aspx



# 中国古代诗歌生成

http://jiuge.thunlp.org/

# 大纲

- 典型任务
- 词表示
- 神经网络进行文本分类
- 神经网络进行机器翻译

# 词表示

- 独热(one-hot)表示

Flower: [0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0]

Phone: [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0]

维度: 词汇表的大小 |*V*| (可能是数百万)

- 问题:
  - 维度太高
  - 没有表达词之间的关系

dist(“kid”, “child”) 与 dist(“flower”, “car”) 没有关系

# 词表示

- 动机：用一个词的邻居来表示这个词

"You shall know a word by the company it keeps"
(J. R. Firth 1957: 11)

…government debt problems turning into banking crises as has happened in…
…saying that Europe needs unified banking regulation to replace the hodgepodge…

这些词表示 *banking*

- 方法：共现矩阵(co-occurrence matrix)

Slide from Richard Socher

# 基于窗口的共现矩阵

- 窗口长度 1 (通常 5-10)
- 对称 (不区分左边或右边)

示例语料

I like deep learning.
I like NLP.
I enjoy flying.

| counts | I | like | deep | learning | NLP | enjoy | flying |
|---|---|---|---|---|---|---|---|
| I | 0 | 2 | 0 | 0 | 0 | 1 | 0 |
| like | 2 | 0 | 1 | 0 | 1 | 0 | 0 |
| deep | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| learning | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| NLP | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| enjoy | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| flying | 0 | 0 | 0 | 0 | 0 | 1 | 0 |

Slide from Richard Socher

# 问题

这种共现矩阵表示形式有什么问题？

- 维度太高
- 表示稀疏使模型不够鲁棒
- 在词汇表中添加一个单词时需要重新计算所有单词的表示形式

# 用低维向量表示单词

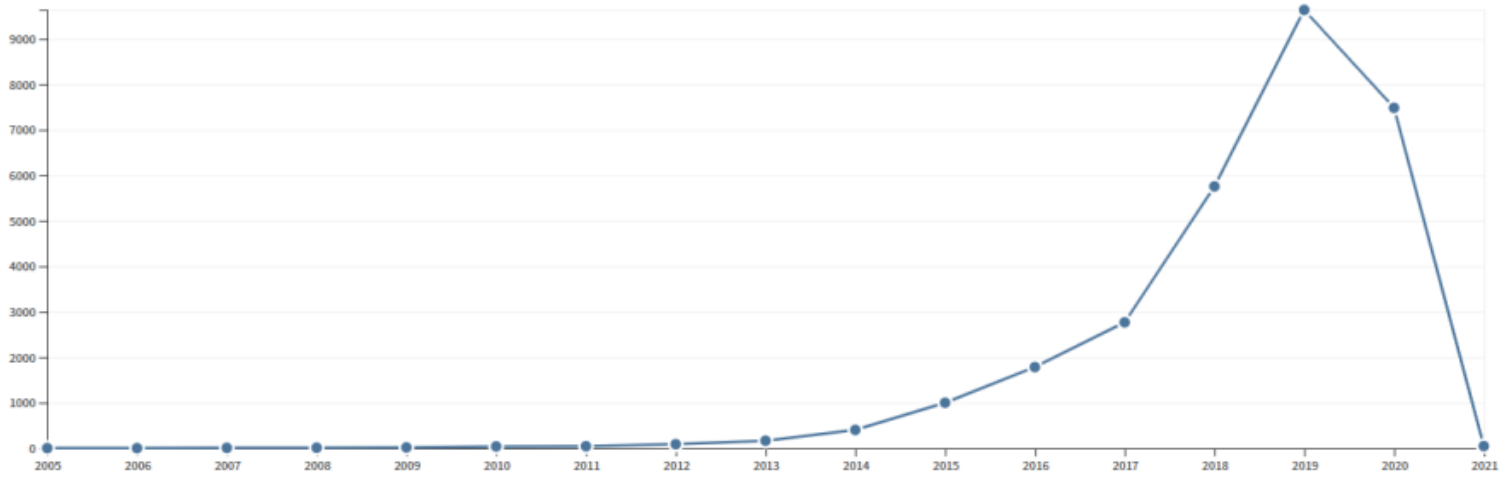
- 目标: 以固定的、低维度的向量存储“大多数”重要信息：
  - 通常约25-1000维
  - 基于这种表示，很容易来执行任务（分类，生成等）

密集向量

- 方法：
  - 神经概率语言模型
    - Bengio et al., A neural probabilistic language model. Journal of Machine Learning Research, 2003.
  - 最近更简单更快的模型: word2vec
    - Mikolov et al., Distributed representations of words and phrases and their compositionality, NeurIPS 2013

# 神经概率语言模型的影响

Bengio et al., A neural probabilistic language model. Journal of Machine Learning Research, 2003

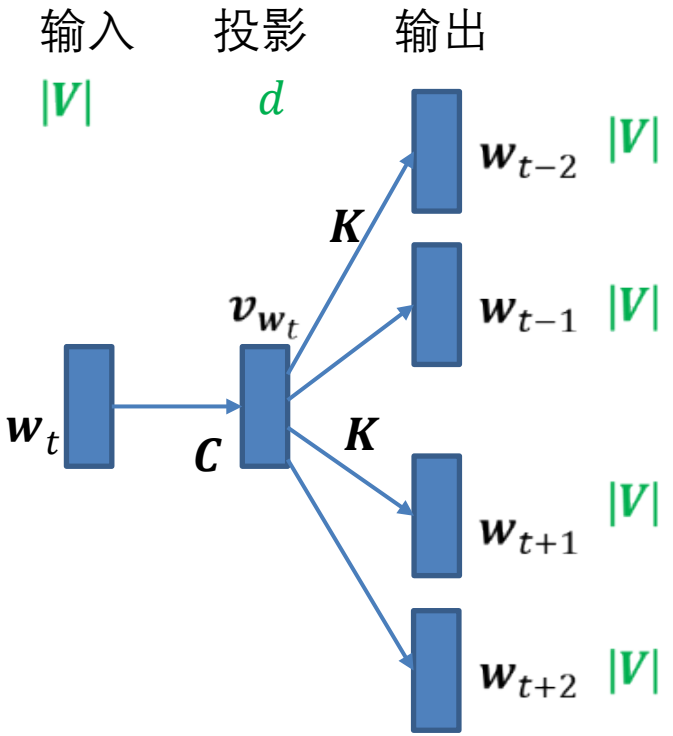


Yoshua Bengio
**Turing Award 2018**

…In 2000 he made a major contribution to natural language processing with the paper “A Neural Probabilistic Language Model.”

# word2vec的主要思想

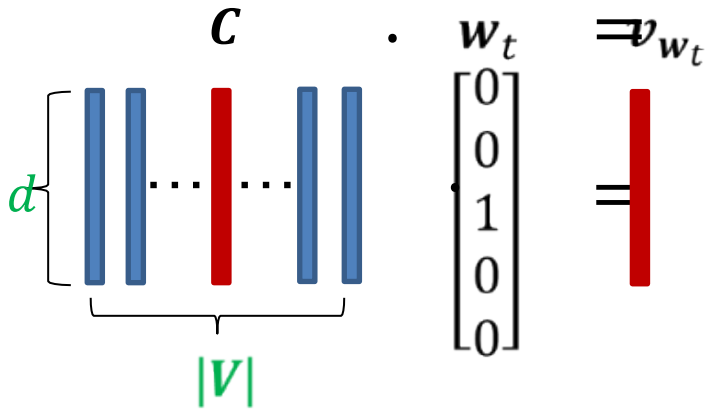
- 不统计词的共现数，而是预测每个词周围的词




$|V|$ 大约$10^5$~$10^7$
$d$ 大约 50~1000

- $w_t \in R^{|V|}$, $v_{w_t} \in R^d$
- $C \in R^{d\times|V|}$: 每列表示词汇表中词的向量

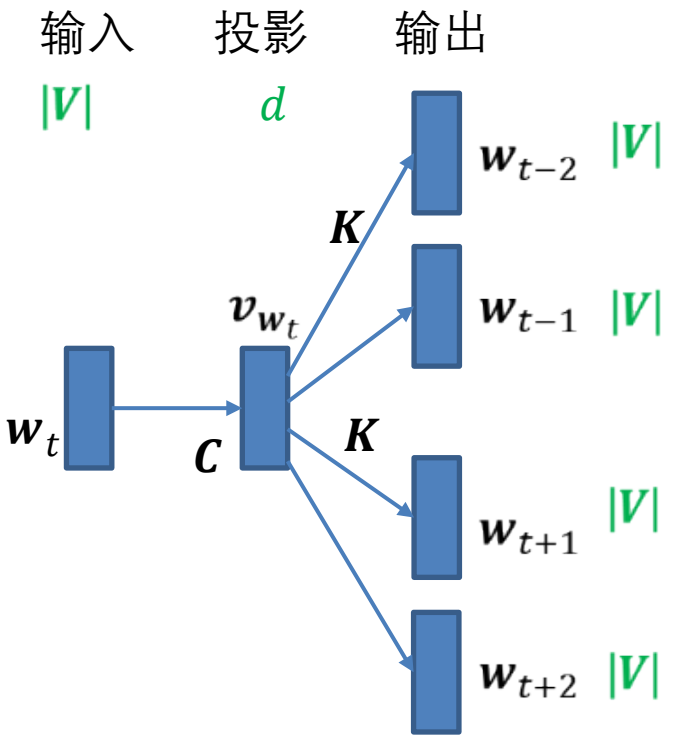
① 将 $w_t$ 通过$C$ 投影到 $v_{w_t}$




独热(one-hot) 向量 $w_t$ 选择 $C$中相应的向量

# word2vec的主要思想

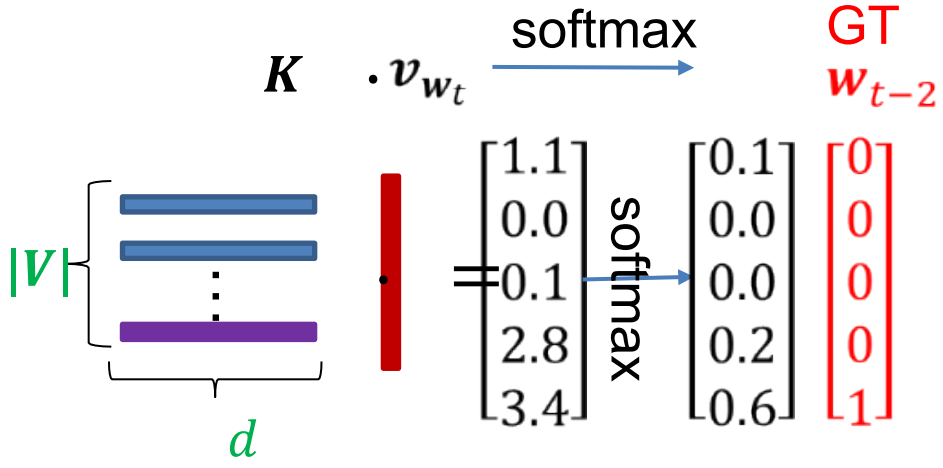
- 不统计词的共现数，而是预测每个词周围的词




$|V|$ 大约$10^5$~$10^7$
$d$ 大约 50~1000

- $w_t \in R^{|V|}$, $v_{w_t} \in R^d$
- $C \in R^{d\times|V|}$:每列表示词汇表中词的向量

② 将 $v_{w_t}$ 通过$K \in R^{|V|\times d}$ 投影到位置为$t-2$的词



# 输入向量与输出向量

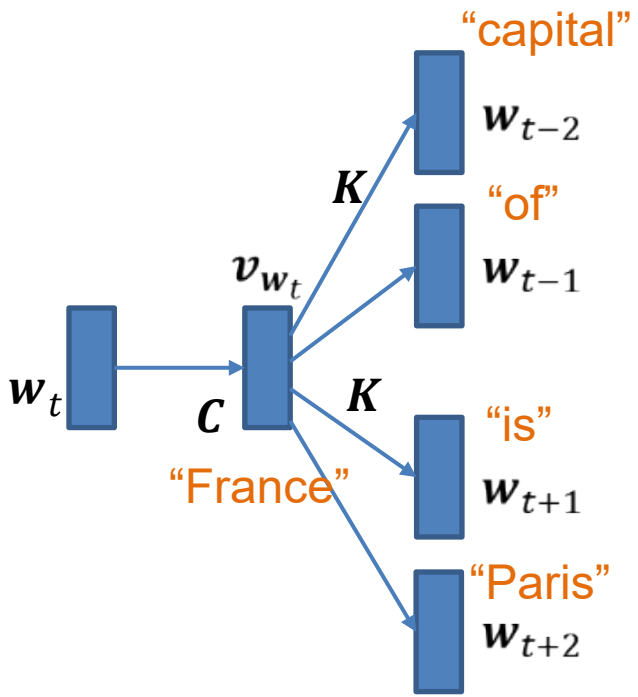
- $\boldsymbol{C}$ 的每一列对应一个词: “输入向量”
- $\boldsymbol{K}$ 的每一行对应一个词: “输出向量”

- 要求:
  - 需要预定义 $\boldsymbol{C}$ 的列与语料库中词之间的对应关系
  - 必须在$\boldsymbol{K}$中使用相同顺序的行

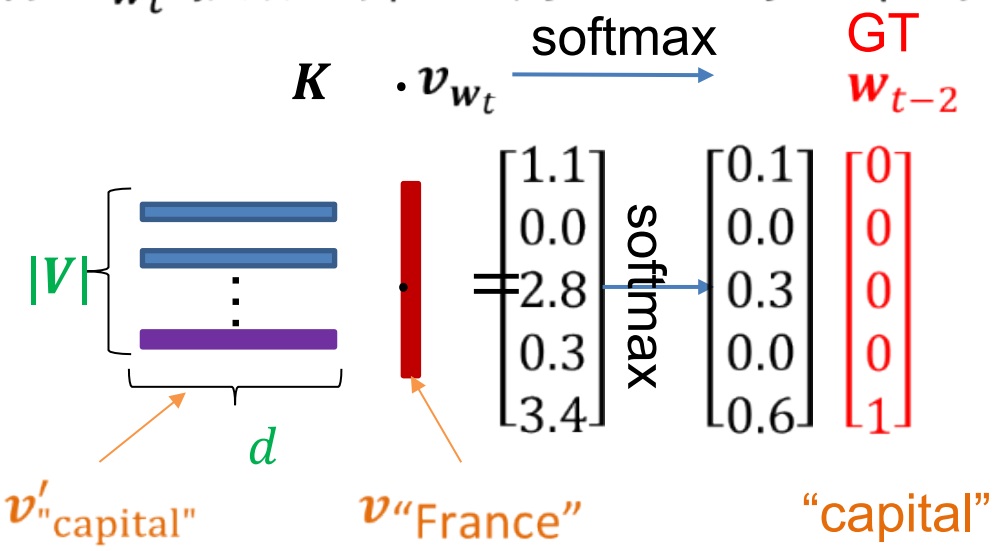
经过学习后，$\boldsymbol{C}$的第$i$列与$\boldsymbol{K}$的第$i$行可以取平均来表示第$i$个词

# word2vec的主要思想

"…capital of France is Paris…"




- 将 $v_{w_t}$ 投影到位置为$t-2$ 的一个词

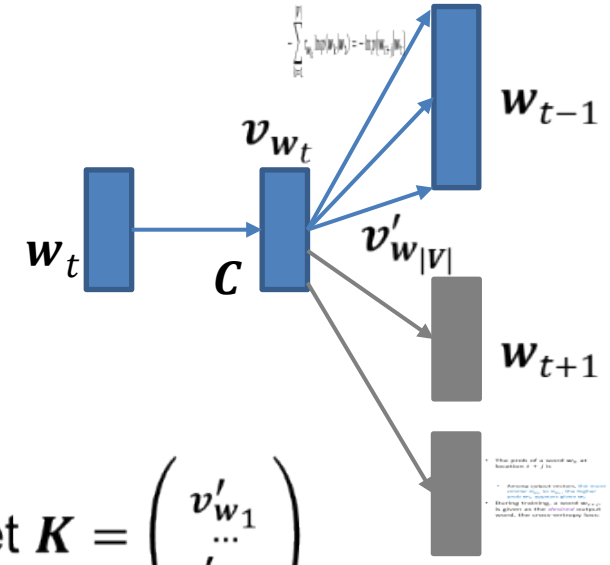



- $K$ 中与"capital"对应的行应当与$v_{"France"}$相似
- 类似地，与"of", "is", "Paris"对应的行也应当与$v_{"France"}$相似

# 公式

- 目标: 给定一个输入词，最大化周围词的概率

skip-gram 模型




Let $K = \begin{pmatrix} v'_{w_1} \\ \cdots \\ v'_{w_{|V|}} \end{pmatrix}$

- 处于位置 $t+j$ $(j=-2,-1,1,2)$ 的词$\boldsymbol{w}_k$的概率为

$$p(\boldsymbol{w}_k|\boldsymbol{w}_t) = \frac{\exp\left(\boldsymbol{v}'_{\boldsymbol{w}_k}{}^{\top}\boldsymbol{v}_{\boldsymbol{w}_t}\right)}{\sum_{m=1}^{|V|}\exp\left(\boldsymbol{v}'_{\boldsymbol{w}_m}{}^{\top}\boldsymbol{v}_{\boldsymbol{w}_t}\right)}$$

- 训练过程中，作为期望输出的词 $\boldsymbol{w}_{t+j}$的交叉熵损失为:

$$-\sum_{k=1}^{|V|} r_{\boldsymbol{w}_k} \ln p(\boldsymbol{w}_k|\boldsymbol{w}_t) = -\ln p(\boldsymbol{w}_{t+j}|\boldsymbol{w}_t)$$

# 目标函数

- 我们在窗口的其他位置有同样的要求。对所有位置的损失求和

$$J^t(\boldsymbol{\theta}) = -\sum_{-c\le j\le c, j\ne 0} \ln p(\boldsymbol{w}_{t+j}|\boldsymbol{w}_t)$$

其中 $c$ 是窗大小，$\boldsymbol{\theta}$ 表示所有参数

- 在所有给定输入 $\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3, \ldots, \boldsymbol{w}_T$ 上求平均

$$J(\boldsymbol{\theta}) = \frac{1}{T}\sum_{t=1}^{T} J^t(\boldsymbol{\theta}) = -\frac{1}{T}\sum_{t=1}^{T}\sum_{-c\le j\le c, j\ne 0} \ln p(\boldsymbol{w}_{t+j}|\boldsymbol{w}_t)$$

- 相当于最大化平均对数概率
- 用反向传播算法和SGD训练模型

# softmax的替代方法

- 在窗内的每一个位置，softmax的输出为

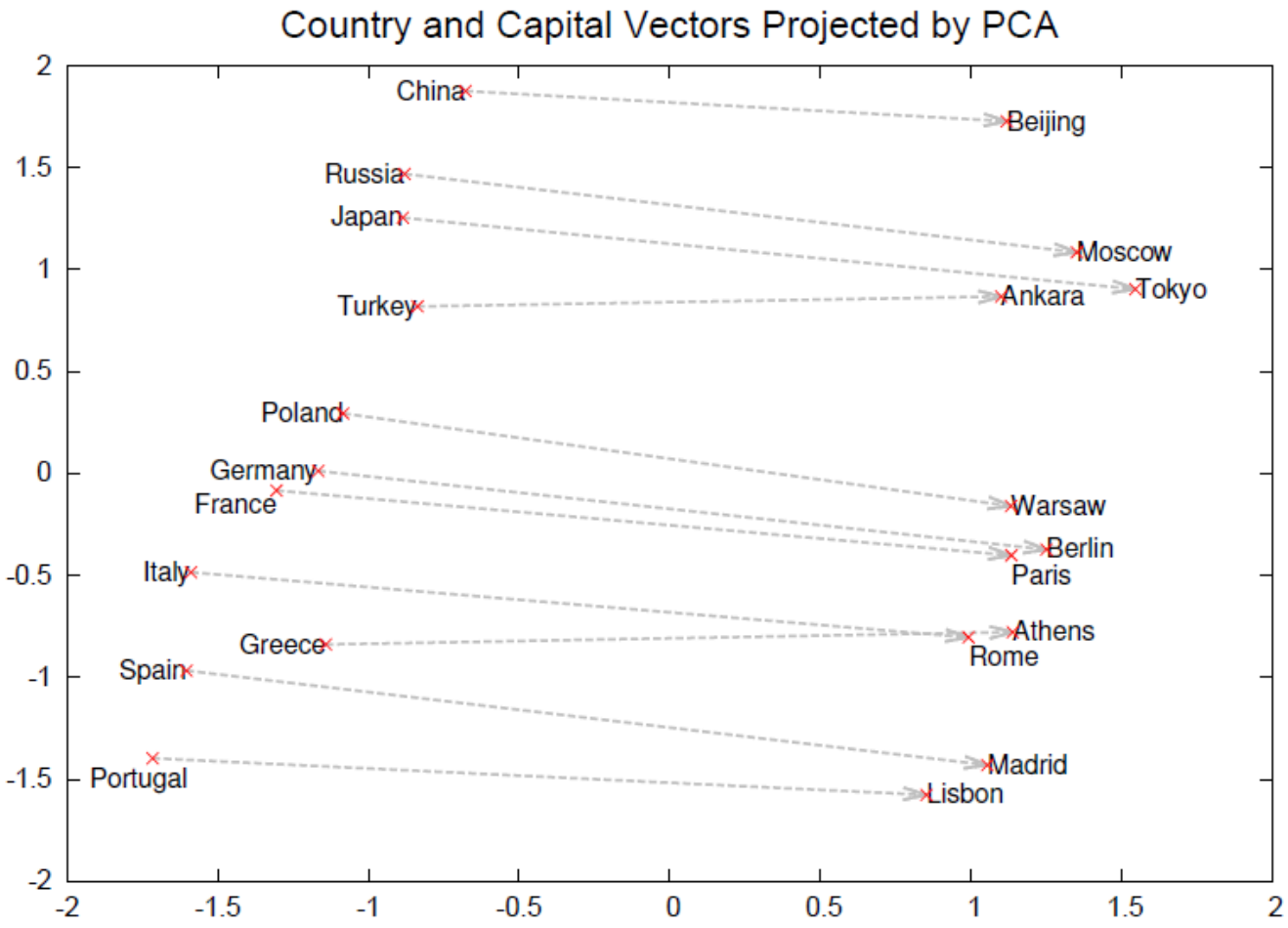
$$p(\boldsymbol{w}_k = \text{"at this location"} | \boldsymbol{w}_t) = \frac{\exp\left(\boldsymbol{v}'_{\boldsymbol{w}_k}{}^{\top} \boldsymbol{v}_{\boldsymbol{w}_t}\right)}{\sum_{m=1}^{|\boldsymbol{V}|} \exp\left(\boldsymbol{v}'_{\boldsymbol{w}_m}{}^{\top} \boldsymbol{v}_{\boldsymbol{w}_t}\right)}$$

其中 $k = 1, \ldots, |\boldsymbol{V}|$

- 这种方法存在什么问题吗？
  - 归一化项的计算耗时长
- 解决方法
  - 层次化 softmax (Morin, Bengio, 2005; Mikolov et al., 2013)
  - 负采样(Negative sampling) (Mikolov et al., 2013)

# 结果

1000维Skip-gram向量的2D PCA投影




Mikolov et al., 2013

# 结果

| Czech + currency | Vietnam + capital | German + airlines | Russian + river | French + actress |
|---|---|---|---|---|
| koruna | Hanoi | airline Lufthansa | Moscow | Juliette Binoche |
| Check crown | Ho Chi Minh City | carrier Lufthansa | Volga River | Vanessa Paradis |
| Polish zolty | Viet Nam | flag carrier Lufthansa | upriver | Charlotte Gainsbourg |
| CTK | Vietnamese | Lufthansa | Russia | Cecile De |

Table 5: Vector compositionality using element-wise addition. Four closest tokens to the sum of two vectors are shown, using the best Skip-gram model.

Mikolov et al., 2013

# 大纲

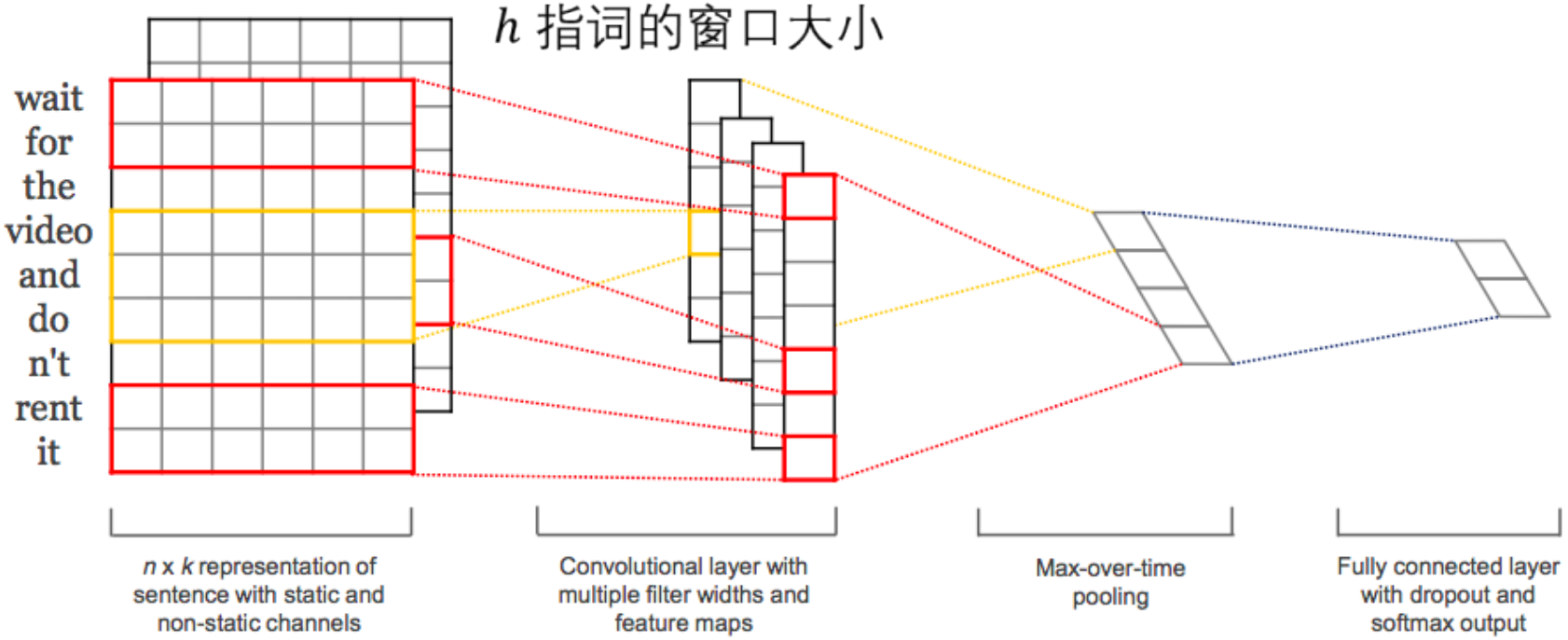
- 典型任务
- 词表示
- 神经网络进行文本分类
- 神经网络进行机器翻译

# 问题

如何表示一个句子或一个段落？

# 神经网络进行文本分类

每个核维度为 $h \times k$, 其中 $h$ 指词的窗口大小




每个词有一个$k$-D 的实向量， $n$个词连接在一起

每个特征向量都是对应于不同$h$的结果

Kim, 2014

# 简单的想法，巨大的影响力

Yoon Kim

FOLLOW

MIT-IBM Watson AI Lab
Verified email at seas.harvard.edu - Homepage
Machine Learning Natural Language Processing Deep Learning

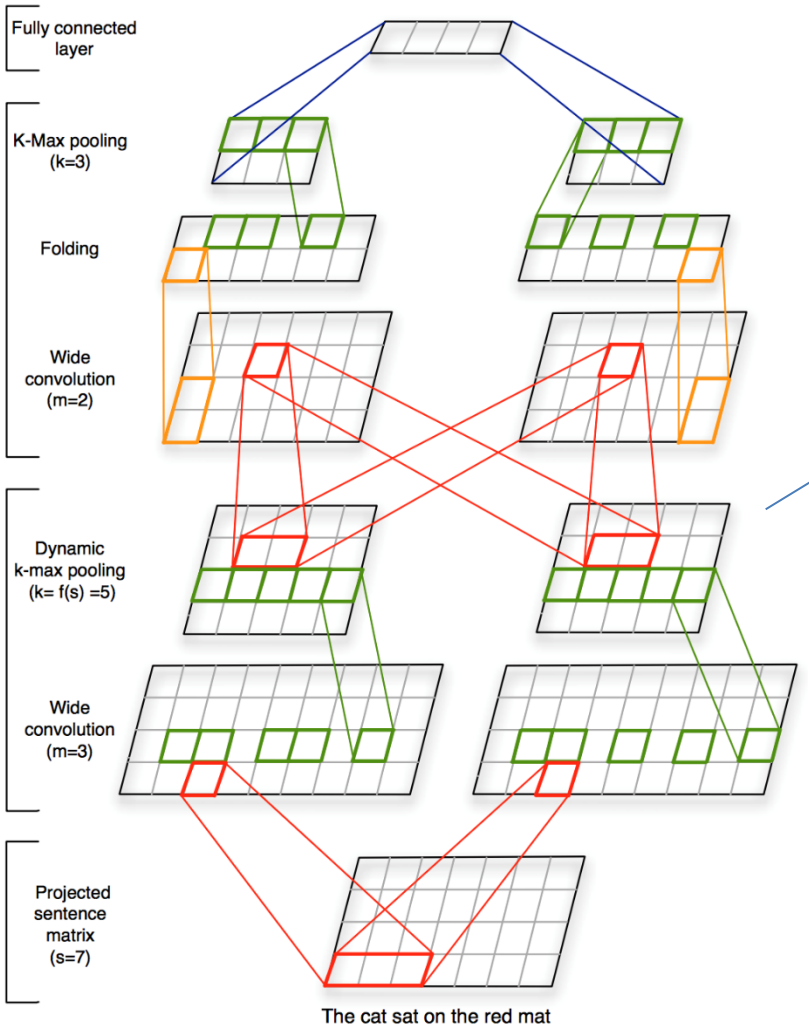
| TITLE | CITED BY | YEAR |
|---|---|---|
| Convolutional Neural Networks for Sentence Classification<br>Y Kim<br>EMNLP 2014 | 9129 | 2014 |
| Character-Aware Neural Language Models<br>Y Kim, Y Jernite, D Sontag, AM Rush<br>AAAI 2016 | 1426 | 2015 |
| OpenNMT: Open-Source Toolkit for Neural Machine Translation<br>G Klein, Y Kim, Y Deng, J Senellart, AM Rush<br>ACL 2017 (System Demonstrations) | 1135 | 2017 |
| Sequence-Level Knowledge Distillation<br>Y Kim, AM Rush<br>EMNLP 2016 | 349 | 2016 |

# 这个模型有什么问题?

- 深度不够
  - 一层卷积和一层池化(pooling)
- 特征不够多样化
  - 每个卷积核会产生一个1维的特征图，即一个特征向量
  - 在一个特征向量上应用全局最大池化(max pooling)，即对所有时间做最大池化，这会产生一个标量

把模型做深的主要困难在哪里?

# 一个更深的模型

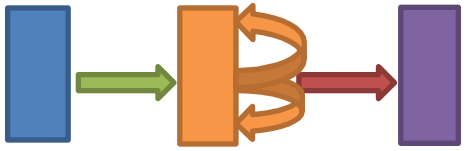



Kalchbrenner et al., 2014

- 沿时间做K-最大池化
  - 解决时间维不等长问题
- 折叠: 两行的元素求和
  - 一种平均池化

# RNN进行文本分类

Elman 网络展开



输入 $\boldsymbol{x}$　隐层 $\boldsymbol{h}$　输出 $\boldsymbol{y}$

$$\boldsymbol{h}(t) = \sigma_h(\boldsymbol{W}_h\boldsymbol{x}(t) + \boldsymbol{U}_h\boldsymbol{h}(t-1) + \boldsymbol{b}_h)$$

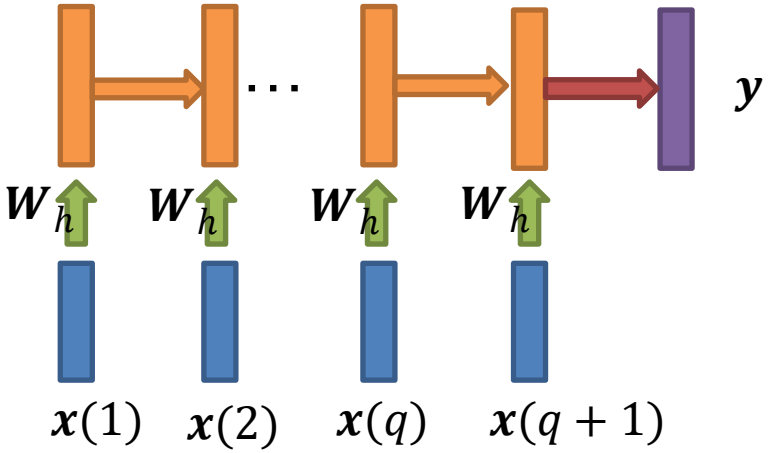
$$\boldsymbol{y}(t) = \sigma_y(\boldsymbol{W}_y\boldsymbol{h}(t) + \boldsymbol{b}_y)$$

- 可以使用 LSTM 或者 GRU
- 可以使用双向 RNN

Deep learning is so cool.

Deep learning is too bad!

- $\boldsymbol{x}$ 随时间变化
- 标签 $\boldsymbol{r}$ 仅在最后一步出现




Pos.

Neg.

# 大纲

- 典型任务
- 词表示
- 神经网络进行文本分类
- 神经网络进行机器翻译

Modified from Abigail See‘s slides (Stanford University)

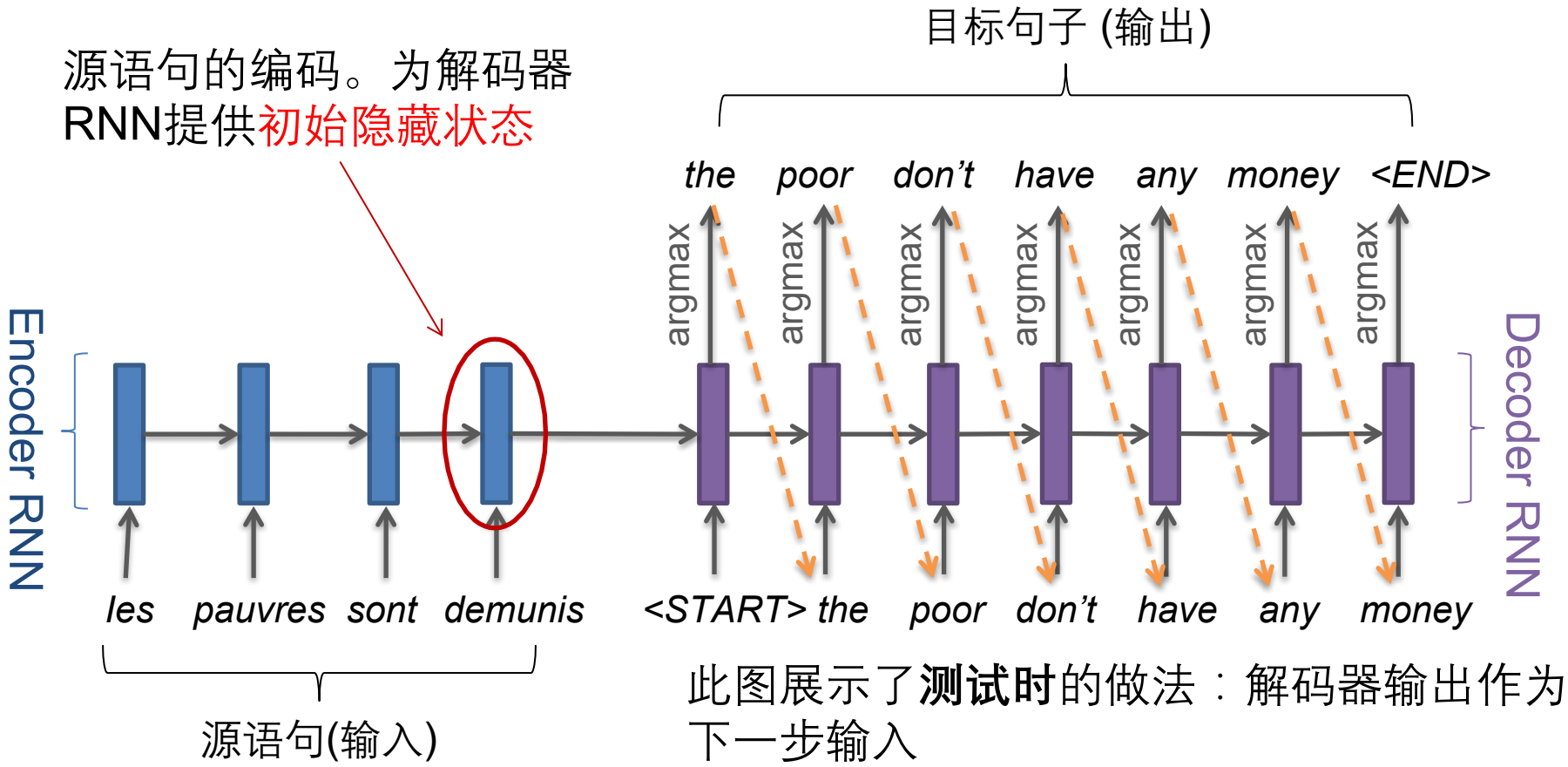
# 序列到序列学习

序列 1 → 序列 2

- 通常会涉及两个RNN: 编码器(encoder)和解码器(decoder)
- 许多NLP任务可以表述为序列到序列:
  - 机器翻译 (法语 → 英语)
  - 总结 (长文本 → 短文本)
  - 对话 (先前的话语 → 接下来的话语)
  - 代码生成 (自然语言 → Python 代码)
  - 旋律产生 (一个乐句 → 下一个乐句)
  - 语音识别 (声音 → 文本)

# 机器翻译 (NMT)

Sutskever et al., 2014




此图展示了**测试时**的做法：解码器输出作为下一步输入

- 编码器RNN 产生一个源语句的编码
- 解码器RNN生成以编码为条件的目标句子

# 编码器和解码器

- 对于编码器，可以使用预训练词向量，例如word2vec，或者one-hot形式词向量
- 对于解码器，使用one-hot形式表示输出

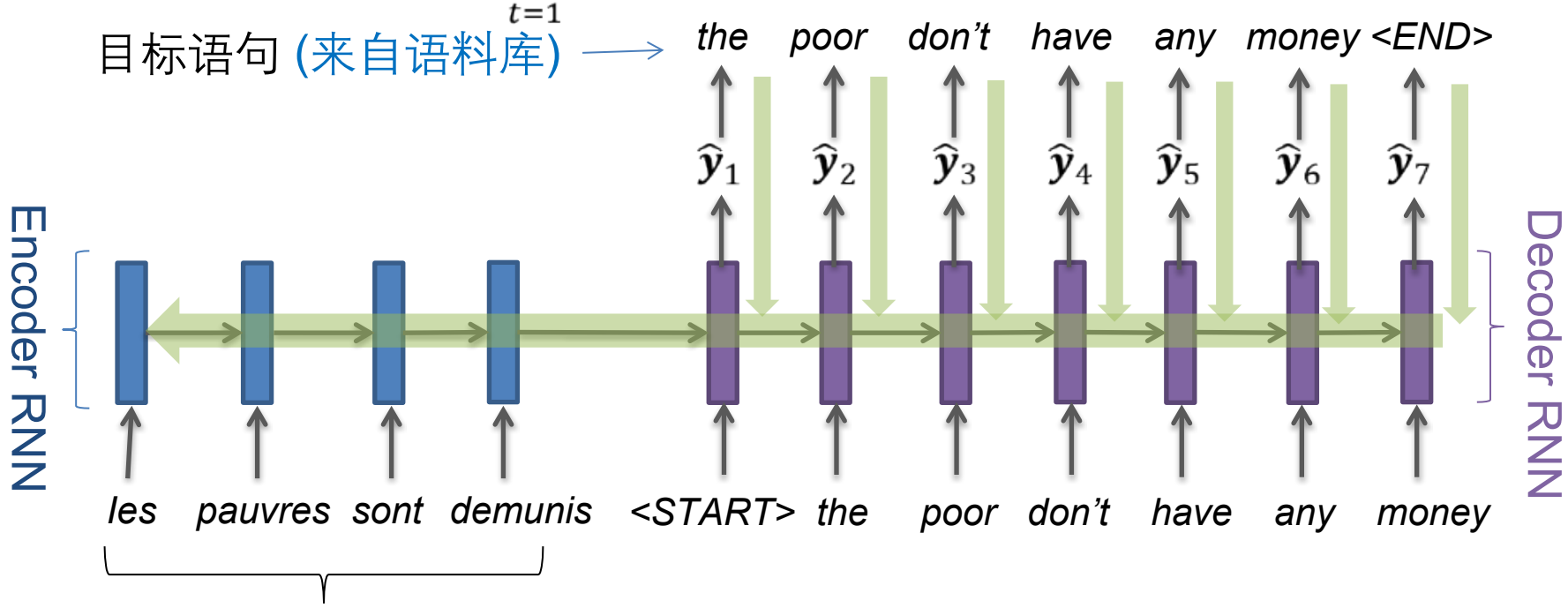
Why?

- 编码器和解码器的字典
  - 对于一些任务，例如，机器翻译，它们是不同的
  - 对于其他任务，例如，总结和对话任务，它们是相同的
- 编码RNN和解码RNN通常是不同的，可以使用深度RNN

# 训练 NMT 系统

= negative log prob of "the" = negative log prob of "have" = negative log prob of <END>

$$J = \frac{1}{T}\sum_{t=1}^{T} J_t = J_1 + J_2 + J_3 + J_4 + J_5 + J_6 + J_7$$

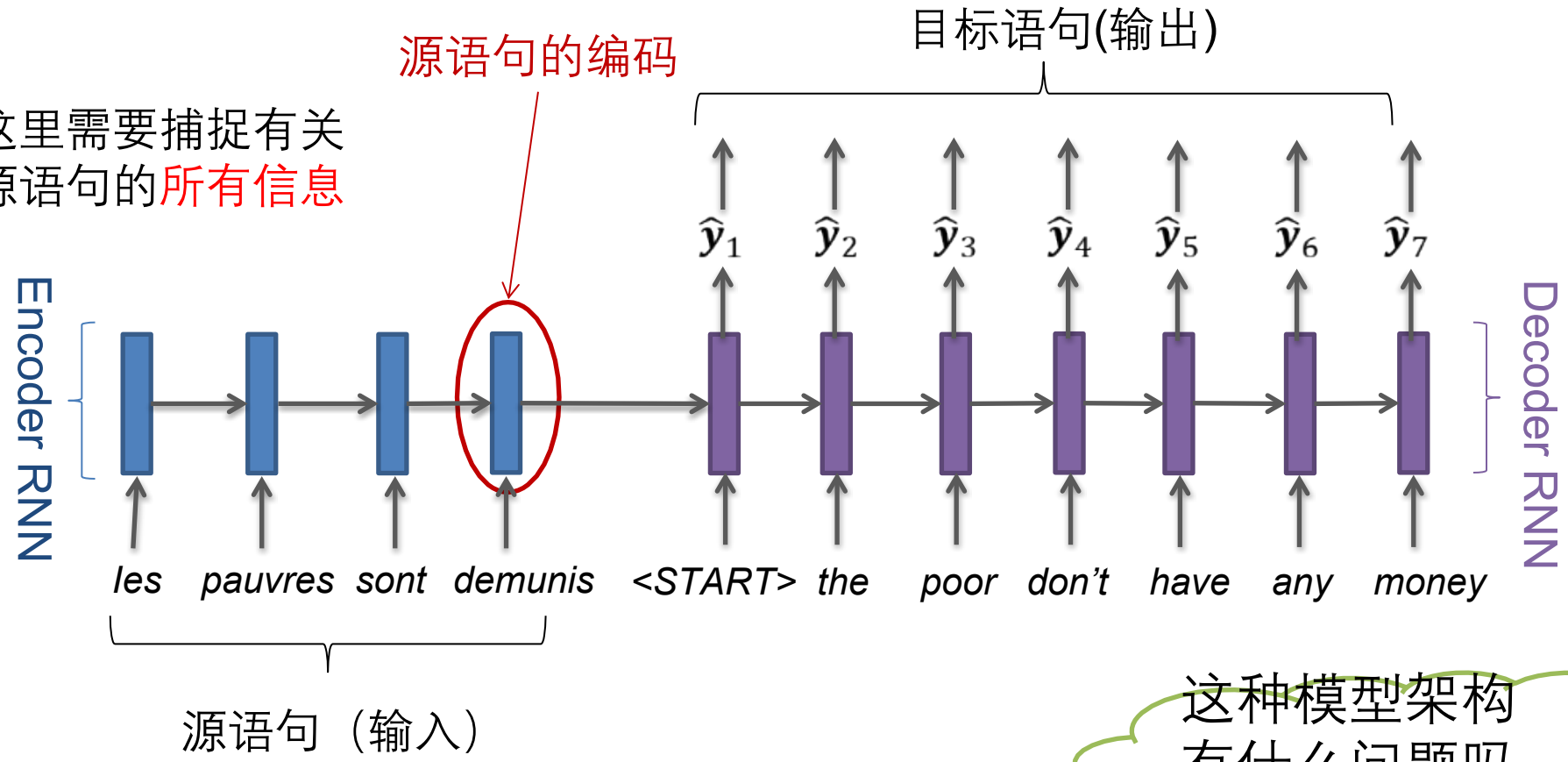



反向传播

Seq2seq 作为**单个系统**进行优化

# 瓶颈问题

这里需要捕捉有关源语句的所有信息



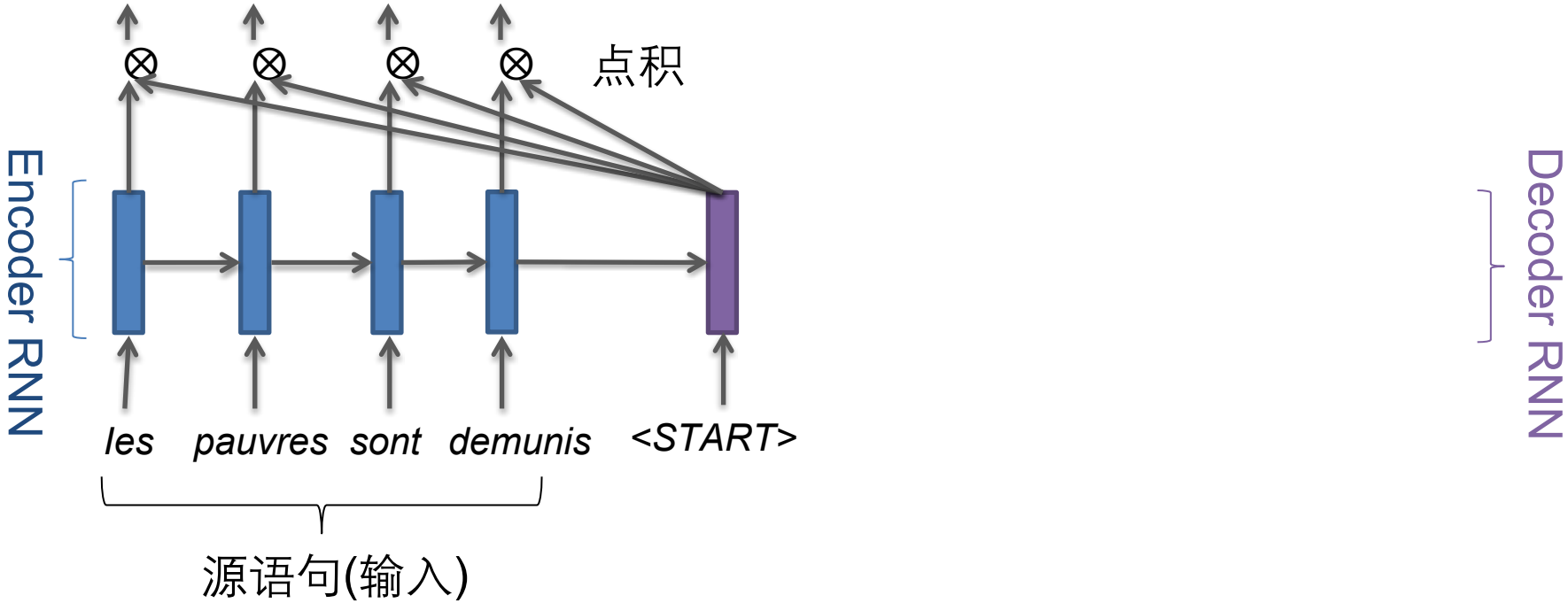

这种模型架构有什么问题吗？

# 注意力机制(Attention)

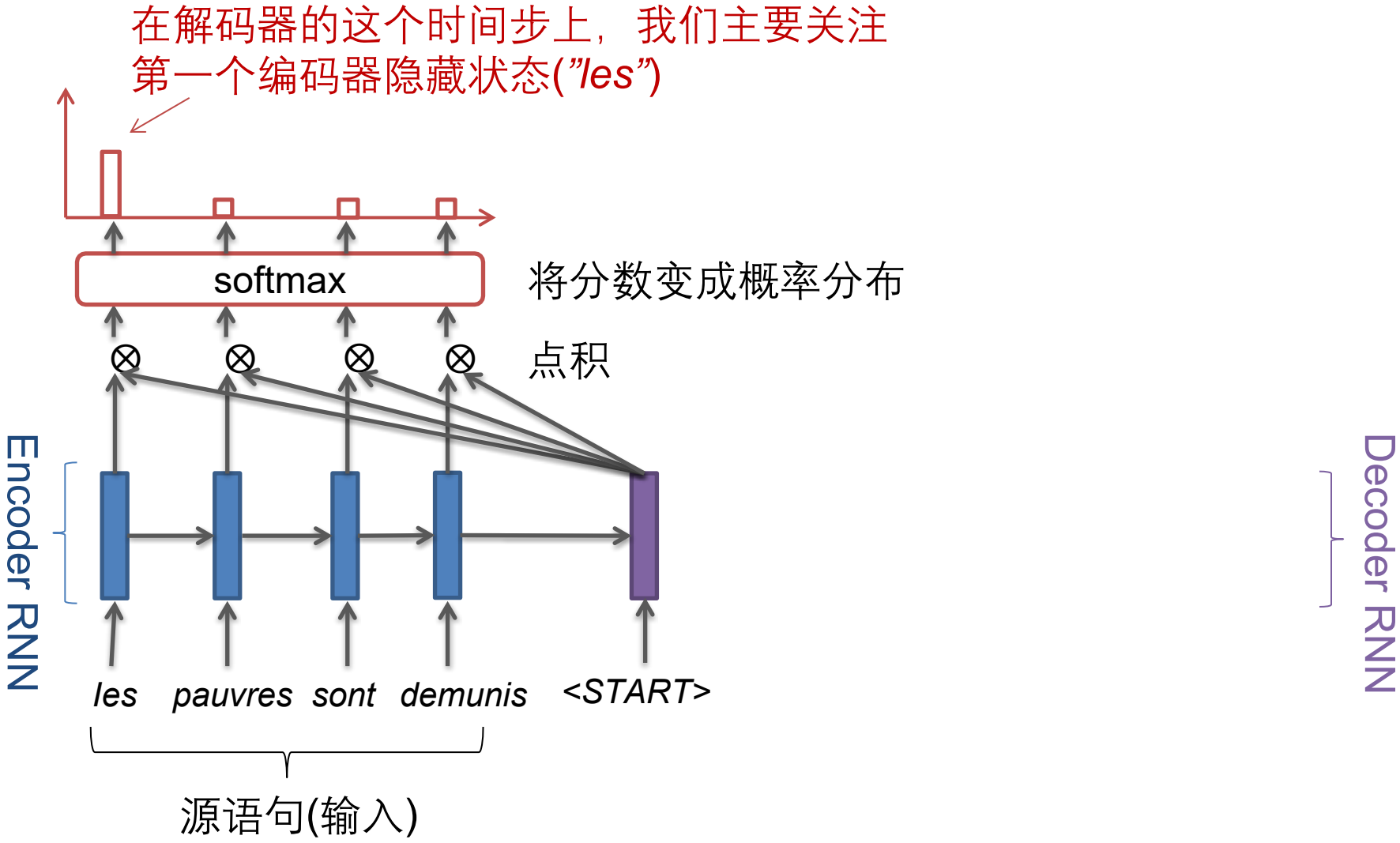
- 注意力机制提供了一个解决瓶颈问题的方法.
- 核心思想: 解码器的每一步中，关注于源序列的特定部分

- 注意力机制有很多形式
- 下面将展示一个例子

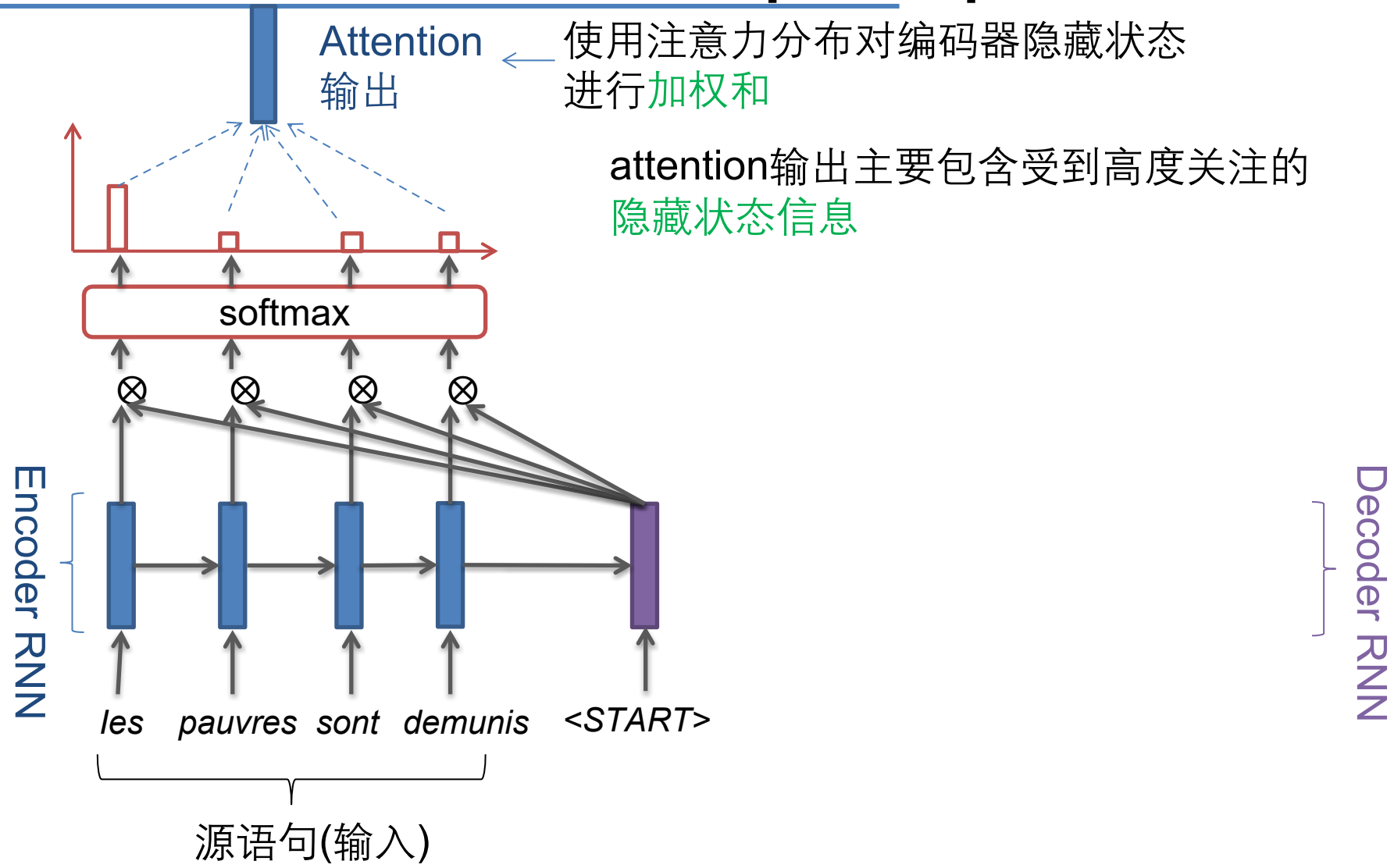
# 带有attention的Seq2seq



# 带有attention的Seq2seq



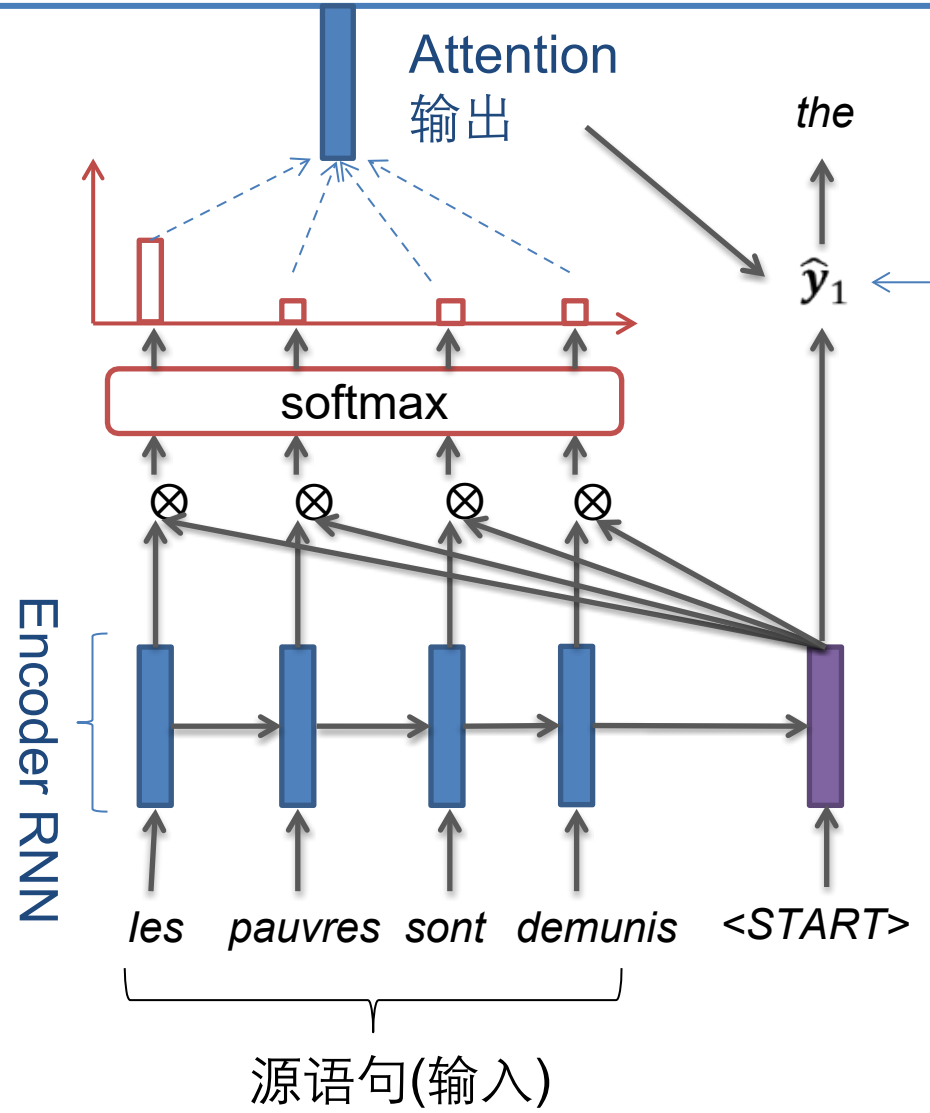
# 带有attention的Seq2seq




使用注意力分布对编码器隐藏状态进行加权和

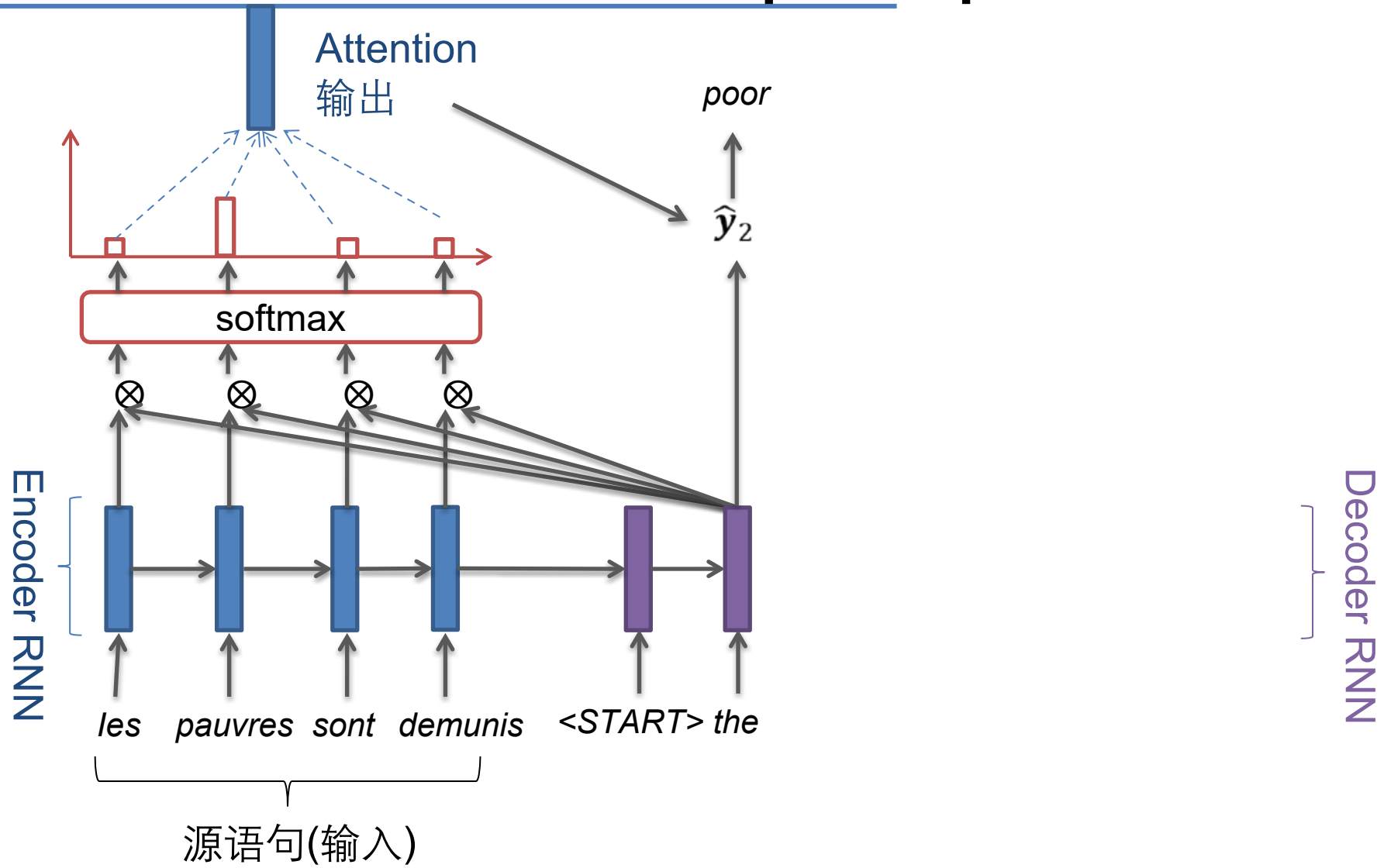
attention输出主要包含受到高度关注的隐藏状态信息

# 带有attention的Seq2seq

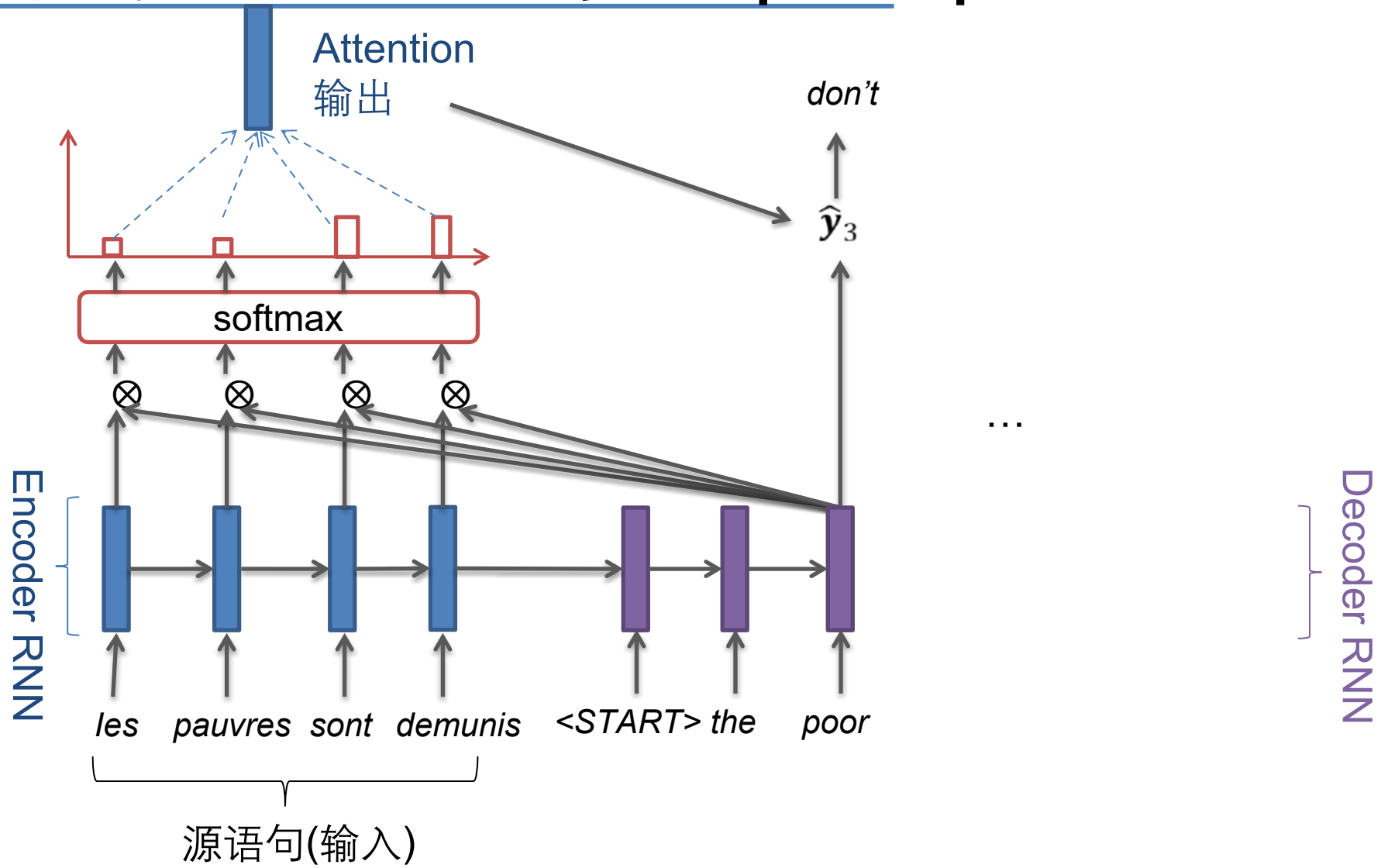



将注意力输出与解码器隐藏状态连接起来，然后像之前一样用于计算 $\hat{y}_1$

# 带有attention的Seq2seq



# 带有attention的Seq2seq



# 公式

编码器的隐藏状态: $\boldsymbol{h}_1, \ldots, \boldsymbol{h}_N \in R^h$
在时间步 $t$, 解码器的隐藏状态 $\boldsymbol{s}_t \in R^h$

- 我们获得此步的注意力得分$\boldsymbol{e}^t$：

$$\boldsymbol{e}^t = [\boldsymbol{s}_t^\top \boldsymbol{h}_1, \ldots, \boldsymbol{s}_t^\top \boldsymbol{h}_N] \in R^N$$

- 应用 softmax 来得到这步的注意力分布

$$\boldsymbol{\alpha}^t = \mathrm{softmax}(\boldsymbol{e}^t) \in R^N$$

- 计算注意力输出

$$\boldsymbol{a}_t = \sum_{i=1}^{N} \alpha_i^t \boldsymbol{h}_i \in R^h$$

- 将注意力输出与解码器隐藏状态连接起来，并按照非注意力seq2seq模型进行操作

$$[\boldsymbol{a}_t; \boldsymbol{s}_t] \in R^{2h}$$

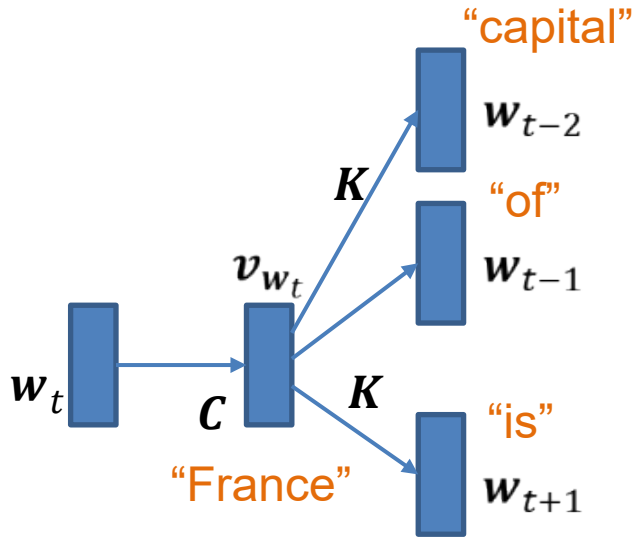
# 总结

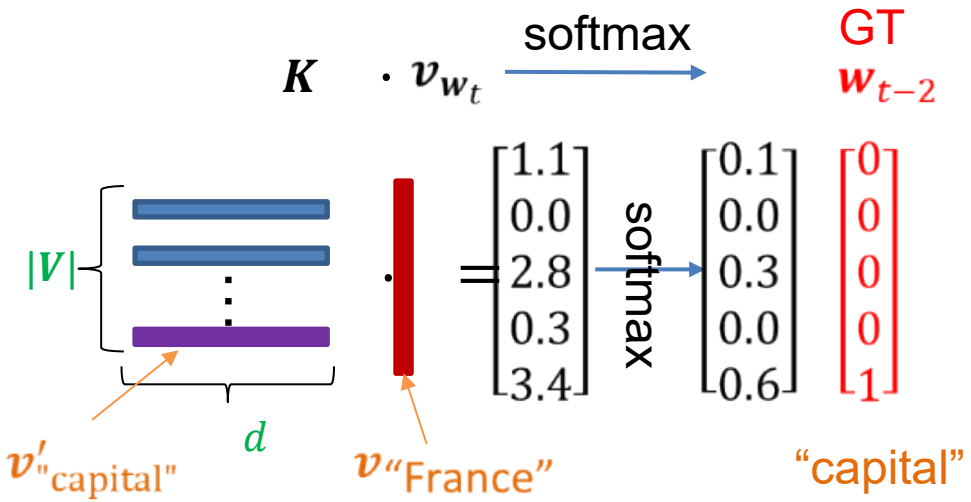
- 典型任务

词性标注和句法分析　问答和对话系统　文本生成

机器翻译　文本/文档分类

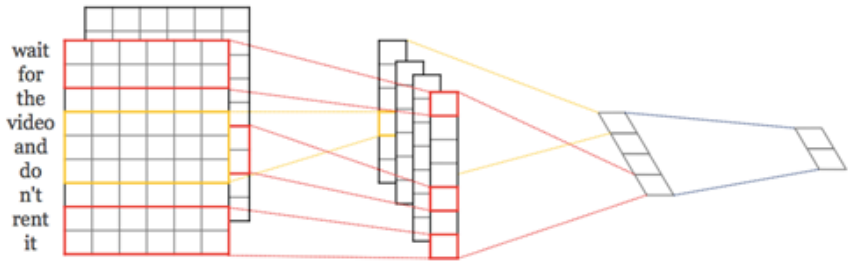
- 词表示

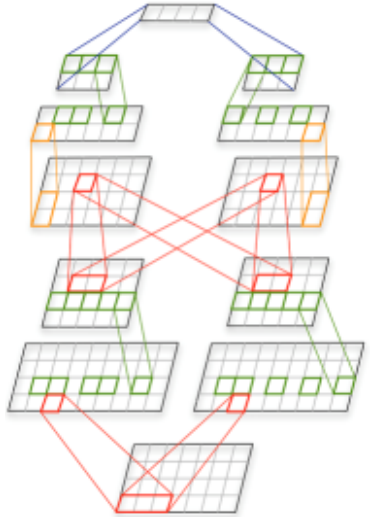


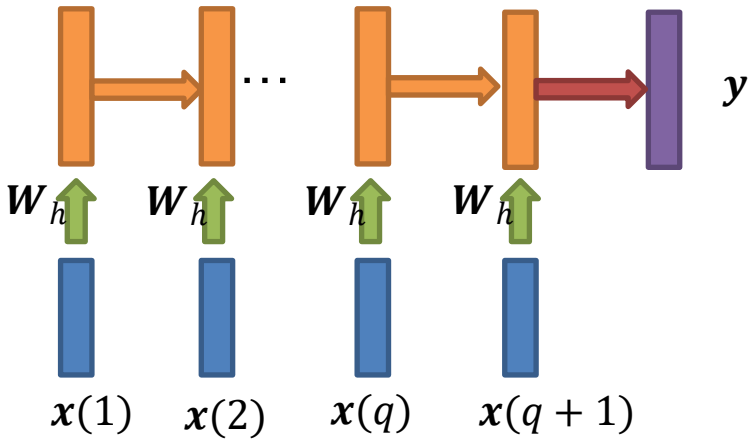



# 总结

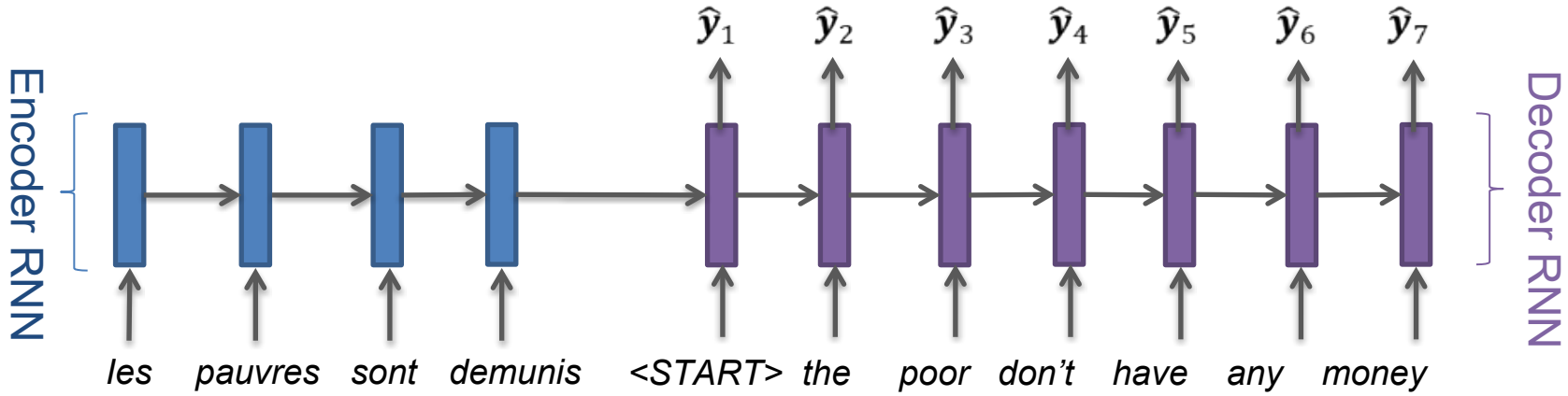
- 神经网络进行文本分类






- 神经网络进行机器翻译



# 主要参考文献


- Kim (2014)
  Convolutional neural networks for sentence classification
  arXiv preprint arXiv:1408.5882
- Bengio, Ducharme, Vincent, Jauvin (2003)
  A neural probabilistic language model
  Journal of Machine Learning Research
- Mikolov, et al. (2013)
  Distributed representations of words and phrases and their compositionality
  NeurIPS
- Sutskever, Vinyals, Le (2014)
  Sequence to Sequence Learning with Neural Networks
  NeurIPS

# 延伸阅读

- Vaswani, Shazeer, Parmar, et al. (2017)
  Attention Is All You Need
  NeurIPS
- Devlin, Chang, Lee, Toutanova (2019)
  BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding
  NAACL